capable of breaking the triple bond and use it for building up their proteins and aminoacids. The process of reduction of gaseous nitrogen (N_2) to ammonia (NH_4) is called Nitrogen fixation. This process is carried out by a group of prokaryotes called diazotrophs.

$$N_2+8H \rightarrow 2\ NH_3 + H_2$$

Cyanobacteria, *Rhizobium* and *Frankia* are some of the examples of **diazotrophs** that can fix atmospheric nitrogen. The fixed ammonia gets incorporated into proteins and amino acids, thus building up organic nitrogen.

$$NH_4 \xrightarrow{\text{assimilation}} \text{Proteins, amino acids}$$

Ammonification

The production of ammonia during the decomposition of organic nitrogen compounds, by micro organisms after the death of plants and animals is called **ammonification** (Figure 11.2). Much of the ammonia released by aerobic decomposition in soil is taken up rapidly by plants and micro organisms and is converted to amino acids.

Bacteria like *Bacillus, Clostridium, Pseudomonas* and fungi like *Aspergillus, Mucor* and *Penicillium* are few examples of micro organisms that can ammonify.

Nitrification

The oxidation of ammonia (NH_3) to nitrate (NO_3) is called **nitrification**. It is carried

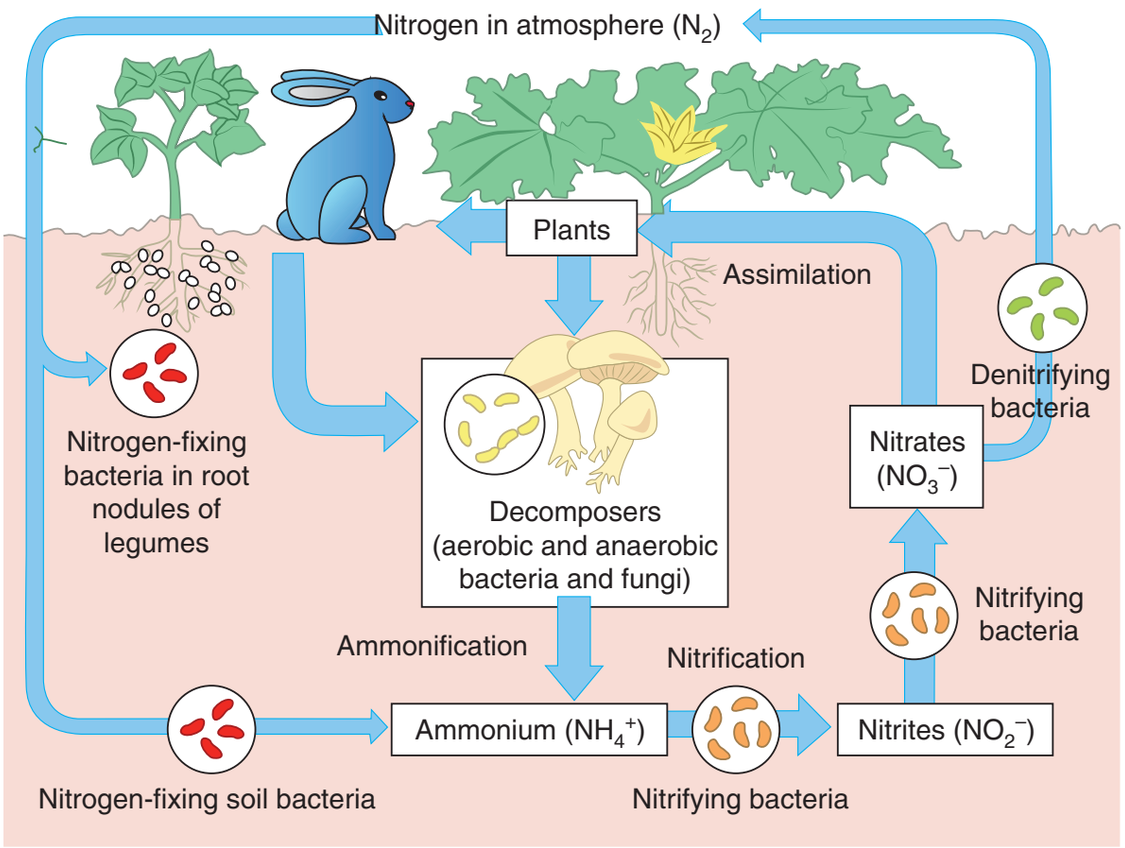


Figure 11.2: Simplified diagram of nitrogen cycle

TN_GOVT_XI_Micro_Biology_CH11.indd 165 16-02-2019 10.04.43 AM